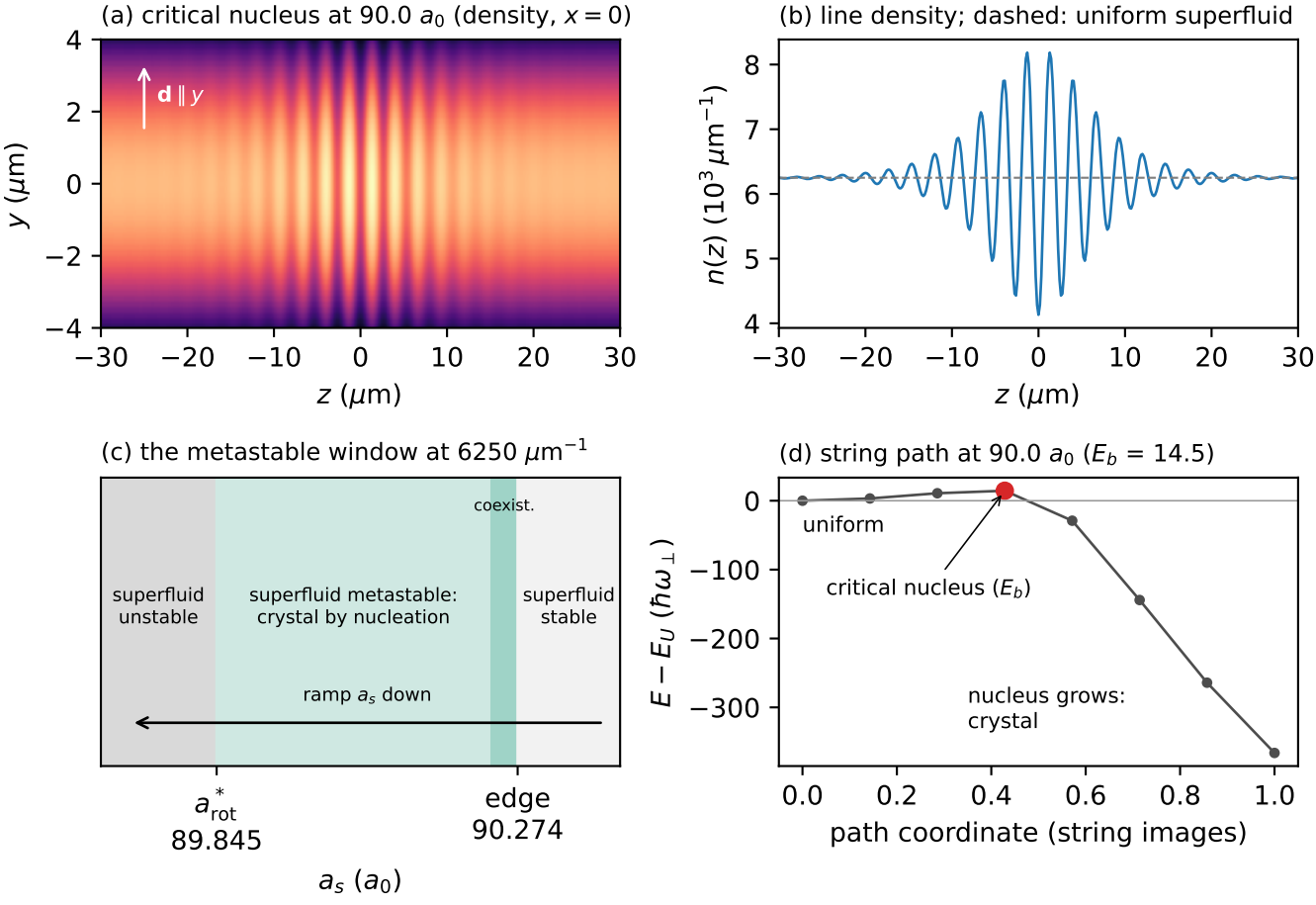

(a) critical nucleus at 90.0 a_0 (density, $x = 0$)
$\mathbf{d} \parallel y$
y (μm)
z (μm)
(b) line density; dashed: uniform superfluid
$n(z)$ ($10^3\,\mu\mathrm{m}^{-1}$)
z (μm)
(c) the metastable window at 6250 $\mu\mathrm{m}^{-1}$
coexist.
superfluid unstable
superfluid metastable: crystal by nucleation
superfluid stable
ramp a_s down
a^*_{rot} 89.845
edge 90.274
a_s (a_0)
(d) string path at 90.0 a_0 (E_b = 14.5)
uniform
critical nucleus (E_b)
nucleus grows: crystal
$E - E_U$ ($\hbar\omega_\perp$)
path coordinate (string images)